

Sprite Compressor Documentation

1. Opening the Tool

Navigate to **Tools > Sprite Compressor** in the Unity Editor menu. The Sprite Compressor window will open.

2. Interface Overview

The window is divided into a few main sections:

- **Header:** Displays the tool title.
- **Drag & Drop Area:** A box where you can drag image files or Texture2D assets to add them to the processing list. Supports **.png**, **.jpg**, and **.jpeg**.
- **Processing Settings:**
 - **Compression Factor (Scale):** A slider (0.1 to 1.0) that determines the target size of the processed sprite. 1.0 is 100% of the original (or trimmed) size, 0.1 is 10%.
 - **Trim Alpha:** A toggle. If enabled, the tool will automatically find and remove fully transparent borders from the sprite before applying the compression factor.
- **Loaded Sprites List:** Displays all images/textures that have been added. Each entry shows the original file name, original dimensions, and the calculated processed dimensions based on the current settings. Each item also has a "Save As..." button and a "Remove" button.
- **Action Buttons:**
 - **Save All as ZIP...**: Processes all sprites in the list with the current settings and saves them into a single ZIP archive.
 - **Clear List:** Removes all sprites from the loaded list after a confirmation prompt.

3. How to Use

1. Open the **Sprite Compressor** window (**Tools > Sprite Compressor**).
2. **Add Sprites/Textures:** Drag image files (from your file explorer) or Texture2D assets (from your Project window) onto the "Drag & Drop Images Here" area. The tool will analyze the files and add supported images to the list.
3. **Adjust Settings:** Modify the **Compression Factor** slider to control the desired output size. Toggle **Trim Alpha** if you want to remove transparent borders. Observe how the "Processed Dimensions" update in the list as you change settings.
4. **Process and Save:**

- To save a single sprite: Click the "Save As..." button next to the item in the list. Choose the save location and file name (defaults to original name + "_compressed.png", saved as PNG).
 - To save all sprites: Click the "Save All as ZIP..." button. Choose the save location and file name for the ZIP archive. The tool will process all listed sprites and pack them into the ZIP file.
5. **Clear List:** Once you are done processing a batch, you can click "Clear List" to empty the window and prepare for new textures.

Important Notes:

- For the **Trim Alpha** feature to work correctly on textures already in your Unity project, the original Texture2D asset must have the **"Read/Write Enabled"** setting checked in its Import Settings.
- All processed sprites/textures are saved in **PNG** format.
- The calculated processed dimensions are rounded up to the nearest multiple of 4.
- The "Save All as ZIP" function creates temporary files during processing before generating the final ZIP archive.